

Recursion Quiz Practice Problems

Note 1. We call a function *recursive* if it calls itself. Every correct recursive function satisfies two laws:

1. in each recursive call, the input parameter somehow gets “smaller”, and
2. there is a *base case* on the “smallest” input value that has no recursion.

A recursive function that does not satisfy both of these laws will run forever. In python, these "forever loops" raise a `RuntimeError`, but they are more generally called a *stack overflow*.

Note 2. Recursion is closely related to *mathematical induction*, which is covered in discrete math (MATH55) and several other courses. At first, most students find recursion very difficult. But eventually, writing recursive functions makes code easier to understand because it is much easier to write proofs about recursive functions than non-recursive functions. In this class, we will focus on runtime proofs using recursion, and the follow on graph algorithms course (CSCI148) covers more detailed proofs.

Problem 3. Write the output of the final command in the following shell script.

```
1 $ cd; rm -rf quiz; mkdir quiz; cd quiz
2 $ cat > foo.py <<EOF
3 def foo(xs):
4     if len(xs) == 0:
5         return 0
6     ret = foo(xs[1:])
7     return xs[0] + ret
8 try:
9     print('foo([1, 2, 3, 4, 5])=',foo([1, 2, 3, 4, 5]))
10 except RuntimeError:
11     print('StackOverflow')
12 EOF
13 $ python3 foo.py
```

Fraction of LLMs with correct answer: 16 / 16 = 1.00

Problem 4. Write the output of the final command in the following shell script.

```
1 $ cd; rm -rf quiz; mkdir quiz; cd quiz
2 $ cat > foo.py <<EOF
3 def foo(xs):
4     ret = foo(xs[1:])
5     if len(xs) == 0:
6         return 0
7     return xs[0] + ret
8 try:
9     print('foo([1, 2, 3, 4, 5])=',foo([1, 2, 3, 4, 5]))
10 except RuntimeError:
11     print('StackOverflow')
12 EOF
13 $ python3 foo.py
```

Fraction of LLMs with correct answer: 16 / 16 = 1.00

Problem 5. Write the output of the final command in the following shell script.

```
1 $ cd; rm -rf quiz; mkdir quiz; cd quiz
2 $ cat > foo.py <<EOF
3 def foo(xs):
4     return foo(xs[1:]) + xs[0]
5 try:
6     print('foo([1, 2, 3, 4, 5])=',foo([1, 2, 3, 4, 5]))
7 except RuntimeError:
8     print('StackOverflow')
9 EOF
10 $ python3 foo.py
```

Fraction of LLMs with correct answer: $13 / 16 = 0.81$

Problem 6. Write the output of the final command in the following shell script.

```
1 $ cd; rm -rf quiz; mkdir quiz; cd quiz
2 $ cat > foo.py <<EOF
3 def foo(xs):
4     if len(xs) == 0:
5         return 0
6     return - foo(xs[1:]) + xs[0]
7 try:
8     print('foo([1, 2, 3, 4, 5])=',foo([1, 2, 3, 4, 5]))
9 except RuntimeError:
10     print('StackOverflow')
11 EOF
12 $ python3 foo.py
```

Fraction of LLMs with correct answer: $12 / 16 = 0.75$

Problem 7. Write the output of the final command in the following shell script.

```
1 $ cd; rm -rf quiz; mkdir quiz; cd quiz
2 $ cat > foo.py <<EOF
3 def foo(xs):
4     if len(xs) == 0:
5         return 0
6     return foo(xs[1:-1]) * xs[-1]
7 try:
8     print('foo([1, 2, 3, 4, 5])=',foo([1, 2, 3, 4, 5]))
9 except RuntimeError:
10     print('StackOverflow')
11 EOF
12 $ python3 foo.py
```

Fraction of LLMs with correct answer: $10 / 16 = 0.62$

Problem 8. Write the output of the final command in the following shell script.

```
1 $ cd; rm -rf quiz; mkdir quiz; cd quiz
2 $ cat > foo.py <<EOF
3 def foo(xs):
4     if len(xs) == 2:
5         return 0
6     return foo(xs[1:-1]) + xs[-1]
7 try:
8     print('foo([1, 2, 3, 4, 5])=',foo([1, 2, 3, 4, 5]))
9 except RuntimeError:
10     print('StackOverflow')
11 EOF
12 $ python3 foo.py
```

Fraction of LLMs with correct answer: $11 / 16 = 0.69$

Problem 9. Write the output of the final command in the following shell script.

```
1 $ cd; rm -rf quiz; mkdir quiz; cd quiz
2 $ cat > foo.py <<EOF
3 def foo(xs):
4     ret = foo(xs[1:-1]) + xs[-1]
5     if len(xs) == 2:
6         return 0
7     return ret
8 try:
9     print('foo([1, 2, 3, 4, 5])=',foo([1, 2, 3, 4, 5]))
10 except RuntimeError:
11     print('StackOverflow')
12 EOF
13 $ python3 foo.py
```

Fraction of LLMs with correct answer: $14 / 16 = 0.88$

Note 10. In python, it is common to define *local functions* for recursion.

Problem 11. Write the output of the final command in the following shell script.

```
1 $ cd; rm -rf quiz; mkdir quiz; cd quiz
2 $ cat > foo.py <<EOF
3 def foo(xs):
4     def go(i):
5         if i == len(xs):
6             return 0
7         return xs[i] + go(i+1)
8     return go(0)
9 try:
10     print('foo([1, 2, 3, 4, 5])=',foo([1, 2, 3, 4, 5]))
11 except RuntimeError:
12     print('StackOverflow')
13 EOF
14 $ python3 foo.py
```

Fraction of LLMs with correct answer: 16 / 16 = 1.00

Problem 12. Write the output of the final command in the following shell script.

```
1 $ cd; rm -rf quiz; mkdir quiz; cd quiz
2 $ cat > foo.py <<EOF
3 def foo(xs):
4     def go(i):
5         if i == -1:
6             return 0
7         return xs[i] + go(i-1)
8     return go(len(xs)-1)
9 try:
10     print('foo([1, 2, 3, 4, 5])=',foo([1, 2, 3, 4, 5]))
11 except RuntimeError:
12     print('StackOverflow')
13 EOF
14 $ python3 foo.py
```

Fraction of LLMs with correct answer: 16 / 16 = 1.00

Problem 13. Write the output of the final command in the following shell script.

```
1 $ cd; rm -rf quiz; mkdir quiz; cd quiz
2 $ cat > foo.py <<EOF
3 def foo(xs):
4     def go(i):
5         if i == 0:
6             return xs[0]
7         return xs[i] + go(i // 2)
8     return go(len(xs)-1)
9 try:
10     print('foo([1, 2, 3, 4, 5])=',foo([1, 2, 3, 4, 5]))
11 except RuntimeError:
12     print('StackOverflow')
13 EOF
14 $ python3 foo.py
```

Fraction of LLMs with correct answer: $10 / 16 = 0.62$

Problem 14. Write the output of the final command in the following shell script.

```
1 $ cd; rm -rf quiz; mkdir quiz; cd quiz
2 $ cat > foo.py <<EOF
3 def foo(xs):
4     if len(xs) == 0:
5         return 0
6     def go(i):
7         return xs[i] + go(i+1)
8     return go(0)
9 try:
10     print('foo([1, 2, 3, 4, 5])=',foo([1, 2, 3, 4, 5]))
11 except RuntimeError:
12     print('StackOverflow')
13 EOF
14 $ python3 foo.py
```

Fraction of LLMs with correct answer: $10 / 16 = 0.62$

Problem 15. Write the output of the final command in the following shell script.

```
1 $ cd; rm -rf quiz; mkdir quiz; cd quiz
2 $ cat > foo.py <<EOF
3 def foo(xs):
4     def go(i, acc):
5         if len(xs) == i:
6             return acc
7         return go(i+1, acc + xs[i])
8     return go(0, 0)
9 try:
10     print('foo([1, 2, 3, 4, 5])=',foo([1, 2, 3, 4, 5]))
11 except RuntimeError:
12     print('StackOverflow')
13 EOF
14 $ python3 foo.py
```

Fraction of LLMs with correct answer: 15 / 16 = 0.94

Problem 16. Write the output of the final command in the following shell script.

```
1 $ cd; rm -rf quiz; mkdir quiz; cd quiz
2 $ cat > foo.py <<EOF
3 def foo(xs):
4     def go(i, acc):
5         if len(xs) == i:
6             return acc
7         return go(i+1, acc + xs[i])
8     return go(1, 2)
9 try:
10     print('foo([1, 2, 3, 4, 5])=',foo([1, 2, 3, 4, 5]))
11 except RuntimeError:
12     print('StackOverflow')
13 EOF
14 $ python3 foo.py
```

Fraction of LLMs with correct answer: 14 / 16 = 0.88

Problem 17. Write the output of the final command in the following shell script.

```
1 $ cd; rm -rf quiz; mkdir quiz; cd quiz
2 $ cat > foo.py <<EOF
3 def foo(xs):
4     def go(i, acc):
5         if len(xs) == i:
6             return acc
7         return go(i+1, - acc + xs[i])
8     return go(0, 0)
9 try:
10     print('foo([1, 2, 3, 4, 5])=',foo([1, 2, 3, 4, 5]))
11 except RuntimeError:
12     print('StackOverflow')
13 EOF
14 $ python3 foo.py
```

Fraction of LLMs with correct answer: 11 / 16 = 0.69

Problem 18. Write the output of the final command in the following shell script.

```
1 $ cd; rm -rf quiz; mkdir quiz; cd quiz
2 $ cat > foo.py <<EOF
3 def foo(xs):
4     if len(xs) == 0:
5         return 0
6     return xs[0] + foo(xs[1:]) + foo(xs[2:])
7 try:
8     print('foo([1, 2, 3])=',foo([1, 2, 3]))
9 except RuntimeError:
10     print('StackOverflow')
11 EOF
12 $ python3 foo.py
```

Fraction of LLMs with correct answer: 10 / 16 = 0.62

Problem 19. Write the output of the final command in the following shell script.

```
1 $ cd; rm -rf quiz; mkdir quiz; cd quiz
2 $ cat > foo.py <<EOF
3 def foo(xs):
4     if len(xs) == 0:
5         return 0
6     ret = xs[0]
7     ret += foo(xs[1:])
8     ret += foo(xs[:-1])
9     return ret
10 try:
11     print('foo([1, 2, 3])=',foo([1, 2, 3]))
12 except RuntimeError:
13     print('StackOverflow')
14 EOF
15 $ python3 foo.py
```

Fraction of LLMs with correct answer: 11 / 16 = 0.69

Problem 20. Write the output of the final command in the following shell script.

```
1 $ cd; rm -rf quiz; mkdir quiz; cd quiz
2 $ cat > foo.py <<EOF
3 def foo(xs):
4     def go(i, acc):
5         if len(xs) <= i:
6             return acc
7         ret = 0
8         ret += go(i+1, acc + xs[i])
9         ret += go(i+2, acc + xs[i])
10        ret += go(i+3, acc + xs[i])
11        return ret
12    return go(0, 0)
13 try:
14     print('foo([1, 2, 3])=',foo([1, 2, 3]))
15 except RuntimeError:
16     print('StackOverflow')
17 EOF
18 $ python3 foo.py
```

Fraction of LLMs with correct answer: 11 / 16 = 0.69

Problem 21. Write the output of the final command in the following shell script.

```
1 $ cd; rm -rf quiz; mkdir quiz; cd quiz
2 $ cat > foo.py <<EOF
3 def foo(xs):
4     def go(i, acc):
5         if len(xs) <= i:
6             return acc
7         ret = 0
8         ret += go(i+1, acc + xs[i])
9         ret += go(i+1, acc + xs[i])
10        return ret
11    return go(0, 0)
12 try:
13     print('foo([1, 2, 3])=', foo([1, 2, 3]))
14 except RuntimeError:
15     print('StackOverflow')
16 EOF
17 $ python3 foo.py
```

Fraction of LLMs with correct answer: 11 / 16 = 0.69

Problem 22. Write the output of the final command in the following shell script.

```
1 $ cd; rm -rf quiz; mkdir quiz; cd quiz
2 $ cat > foo.py <<EOF
3 def foo(xs):
4     if len(xs) == 10:
5         return 1
6     xs.append(1)
7     return 2 * foo(xs)
8 try:
9     print('foo([])=', foo([]))
10 except RuntimeError:
11     print('StackOverflow')
12 EOF
13 $ python3 foo.py
```

Fraction of LLMs with correct answer: 14 / 16 = 0.88

Problem 23. Write the output of the final command in the following shell script.

```
1 $ cd; rm -rf quiz; mkdir quiz; cd quiz
2 $ cat > foo.py <<EOF
3 def foo(xs):
4     if len(xs) == 10:
5         return 1
6     xs.append(1)
7     return 2 * foo(xs)
8 try:
9     print('foo([1, 2, 3])=',foo([1, 2, 3]))
10 except RuntimeError:
11     print('StackOverflow')
12 EOF
13 $ python3 foo.py
```

Fraction of LLMs with correct answer: $13 / 16 = 0.81$

Problem 24. Write the output of the final command in the following shell script.

```
1 $ cd; rm -rf quiz; mkdir quiz; cd quiz
2 $ cat > foo.py <<EOF
3 def foo(xs):
4     xs.append(1)
5     return 2 * foo(xs)
6     if len(xs) == 10:
7         return 1
8 try:
9     print('foo([1, 2, 3])=',foo([1, 2, 3]))
10 except RuntimeError:
11     print('StackOverflow')
12 EOF
13 $ python3 foo.py
```

Fraction of LLMs with correct answer: $15 / 16 = 0.94$

Note 25. The problems below are based on real world interview problems.

Problem 26. Write the output of the final command in the following shell script.

```
1 $ cd; rm -rf quiz; mkdir quiz; cd quiz
2 $ cat > foo.py <<EOF
3 def sequential_search(xs, y):
4     if len(xs) == 0:
5         return False
6     if xs[0] == y:
7         return True
8     return sequential_search(xs[1:], y)
9 try:
10     xs = [1, 3, 5, 4, 2, 0]
11     print('sequential_search(xs, 2)=' , sequential_search(xs, 2))
12 except RuntimeError:
13     print('StackOverflow')
14 EOF
15 $ python3 foo.py
```

Fraction of LLMs with correct answer: 16 / 16 = 1.00

Problem 27. Write the output of the final command in the following shell script.

```
1 $ cd; rm -rf quiz; mkdir quiz; cd quiz
2 $ cat > foo.py <<EOF
3 def sequential_search(xs, y):
4     if len(xs) == 0:
5         return False
6     if xs[-1] == y:
7         return True
8     return sequential_search(xs[1:], y)
9 try:
10     xs = [1, 3, 5, 4, 2, 0]
11     print('sequential_search(xs, 2)=' , sequential_search(xs, 2))
12 except RuntimeError:
13     print('StackOverflow')
14 EOF
15 $ python3 foo.py
```

Fraction of LLMs with correct answer: 11 / 16 = 0.69

Problem 28. Write the output of the final command in the following shell script.

```
1 $ cd; rm -rf quiz; mkdir quiz; cd quiz
2 $ cat > foo.py <<EOF
3 def binary_search(xs, y):
4     if len(xs) == 0:
5         return False
6     mid = len(xs) // 2
7     if xs[mid] > y:
8         return binary_search(xs[:mid], y)
9     if xs[mid] < y:
10        return binary_search(xs[mid+1:], y)
11    return True
12 try:
13     xs = [1, 3, 5, 7, 9]
14     print('binary_search(xs, 9)=', binary_search(xs, 9))
15 except RuntimeError:
16     print('StackOverflow')
17 EOF
18 $ python3 foo.py
```

Fraction of LLMs with correct answer: 16 / 16 = 1.00

Problem 29. Write the output of the final command in the following shell script.

```
1 $ cd; rm -rf quiz; mkdir quiz; cd quiz
2 $ cat > foo.py <<EOF
3 def binary_search(xs, y):
4     if len(xs) == 0:
5         return False
6     mid = len(xs) // 2
7     if xs[mid] >= y:
8         return binary_search(xs[:mid], y)
9     return binary_search(xs[mid+1:], y)
10 try:
11     xs = [1, 3, 5, 7, 9]
12     print('binary_search(xs, 5)=', binary_search(xs, 5))
13 except RuntimeError:
14     print('StackOverflow')
15 EOF
16 $ python3 foo.py
```

Fraction of LLMs with correct answer: 14 / 16 = 0.88

Problem 30. Write the output of the final command in the following shell script.

```
1 $ cd; rm -rf quiz; mkdir quiz; cd quiz
2 $ cat > foo.py <<EOF
3 def binary_search(xs, y):
4     if len(xs) == 0:
5         return False
6     mid = len(xs) // 2
7     if xs[mid] > y:
8         return binary_search(xs[:mid], y)
9     if xs[mid] < y:
10        return binary_search(xs[mid:], y)
11    return True
12 try:
13     xs = [1, 3, 5, 7, 9]
14     print('binary_search(xs, 11)=', binary_search(xs, 11))
15 except RuntimeError:
16     print('StackOverflow')
17 EOF
18 $ python3 foo.py
```

Fraction of LLMs with correct answer: 14 / 16 = 0.88

Problem 31. Write the output of the final command in the following shell script.

```
1 $ cd; rm -rf quiz; mkdir quiz; cd quiz
2 $ cat > foo.py <<EOF
3 def binary_search(xs, y):
4     if len(xs) == 0:
5         return False
6     mid = len(xs) // 2
7     if xs[mid] > y:
8         return binary_search(xs[:mid-1], y)
9     if xs[mid] < y:
10        return binary_search(xs[mid+1:], y)
11    return True
12 try:
13     xs = [1, 3, 5, 7, 9]
14     print('binary_search(xs, 3)=', binary_search(xs, 3))
15 except RuntimeError:
16     print('StackOverflow')
17 EOF
18 $ python3 foo.py
```

Fraction of LLMs with correct answer: 14 / 16 = 0.88

Problem 32. Write the output of the final command in the following shell script.

```
1 $ cd; rm -rf quiz; mkdir quiz; cd quiz
2 $ cat > foo.py <<EOF
3 def trinary_search(xs, y):
4     if len(xs) == 0:
5         return False
6     mid1 = len(xs) // 3
7     mid2 = 2 * len(xs) // 3
8     if xs[mid1] == y:
9         return True
10    if xs[mid2] == y:
11        return True
12    if y < xs[mid1]:
13        return trinary_search(xs[:mid1], y)
14    if y < xs[mid2]:
15        return trinary_search(xs[mid1+1:mid2], y)
16    return trinary_search(xs[mid2+1:], y)
17 try:
18     xs = [1, 3, 5, 7, 9, 11]
19     print('trinary_search(xs, 7)=',trinary_search(xs, 7))
20 except RuntimeError:
21     print('StackOverflow')
22 EOF
23 $ python3 foo.py
```

Fraction of LLMs with correct answer: 14 / 16 = 0.88

Problem 33. Write the output of the final command in the following shell script.

```
1 $ cd; rm -rf quiz; mkdir quiz; cd quiz
2 $ cat > foo.py <<EOF
3 def trinary_search(xs, y):
4     if len(xs) == 0:
5         return False
6     mid1 = len(xs) // 3
7     mid2 = 2 * len(xs) // 3
8     if y == xs[mid1]:
9         return True
10    if y < xs[mid1]:
11        return trinary_search(xs[:mid1], y)
12    if y < xs[mid2]:
13        return trinary_search(xs[mid1+1:mid2], y)
14    return trinary_search(xs[mid2:], y)
15 try:
16     xs = [1, 3, 5, 7, 9, 11]
17     print('trinary_search(xs, 5)=',trinary_search(xs, 5))
18 except RuntimeError:
19     print('StackOverflow')
20 EOF
21 $ python3 foo.py
```

Fraction of LLMs with correct answer: 14 / 16 = 0.88

Problem 34. Write the output of the final command in the following shell script.

```
1 $ cd; rm -rf quiz; mkdir quiz; cd quiz
2 $ cat > foo.py <<EOF
3 def trinary_search(xs, y):
4     if len(xs) == 0:
5         return False
6     mid1 = len(xs) // 3
7     mid2 = 2 * len(xs) // 3
8     if xs[mid1] >= y:
9         return True
10    if xs[mid2] >= y:
11        return True
12    if y < xs[mid1]:
13        return trinary_search(xs[:mid1], y)
14    if y < xs[mid2]:
15        return trinary_search(xs[mid1+1:mid2], y)
16    return trinary_search(xs[mid2+1:], y)
17 try:
18     xs = [1, 3, 5, 7, 9, 11]
19     print('trinary_search(xs, 2)=',trinary_search(xs, 2))
20 except RuntimeError:
21     print('StackOverflow')
22 EOF
23 $ python3 foo.py
```

Fraction of LLMs with correct answer: 11 / 16 = 0.69

Problem 35. Write the output of the final command in the following shell script.

```
1 $ cd; rm -rf quiz; mkdir quiz; cd quiz
2 $ cat > foo.py <<EOF
3 def sequential_search_go(xs, y):
4     def go(i):
5         if i == len(xs):
6             return False
7         if xs[i] == y:
8             return True
9         return go(i+1)
10    return go(0)
11 try:
12     xs = [1, 3, 5, 4, 2, 0]
13     print('sequential_search_go(xs, 2)=',sequential_search_go(xs, 2))
14 except RuntimeError:
15     print('StackOverflow')
16 EOF
17 $ python3 foo.py
```

Fraction of LLMs with correct answer: 16 / 16 = 1.00

Problem 36. Write the output of the final command in the following shell script.

```
1 $ cd; rm -rf quiz; mkdir quiz; cd quiz
2 $ cat > foo.py <<EOF
3 def binary_search_go(xs, y):
4     def go(lo, hi):
5         if lo > hi:
6             return False
7         mid = (lo + hi) // 2
8         if xs[mid] > y:
9             return go(lo, mid - 1)
10        if xs[mid] < y:
11            return go(mid + 1, hi)
12        return True
13    return go(0, len(xs) - 1)
14 try:
15     xs = [1, 3, 5, 7, 9]
16     print('binary_search_go(xs, 9)=', binary_search_go(xs, 9))
17 except RuntimeError:
18     print('StackOverflow')
19 EOF
20 $ python3 foo.py
```

Fraction of LLMs with correct answer: 16 / 16 = 1.00

Problem 37. Write the output of the final command in the following shell script.

```
1 $ cd; rm -rf quiz; mkdir quiz; cd quiz
2 $ cat > foo.py <<EOF
3 def binary_search_go(xs, y):
4     def go(lo, hi):
5         if lo > hi:
6             return False
7         mid = (lo + hi) // 2
8         if xs[mid] >= y:
9             return go(lo, mid - 1)
10        return go(mid + 1, hi)
11    return go(0, len(xs) - 1)
12 try:
13     xs = [1, 3, 5, 7, 9]
14     print('binary_search_go(xs, 7)=', binary_search_go(xs, 7))
15 except RuntimeError:
16     print('StackOverflow')
17 EOF
18 $ python3 foo.py
```

Fraction of LLMs with correct answer: 14 / 16 = 0.88

Problem 38. Write the output of the final command in the following shell script.

```
1 $ cd; rm -rf quiz; mkdir quiz; cd quiz
2 $ cat > foo.py <<EOF
3 def trinary_search_go(xs, y):
4     def go(lo, hi):
5         if lo > hi:
6             return False
7         mid1 = lo + (hi - lo) // 3
8         mid2 = lo + 2 * (hi - lo) // 3
9         if xs[mid1] == y:
10            return True
11        if xs[mid2] == y:
12            return True
13        if y < xs[mid1]:
14            return go(lo, mid1 - 1)
15        if y < xs[mid2]:
16            return go(mid1 + 1, mid2 - 1)
17        return go(mid2 + 1, hi)
18    return go(0, len(xs) - 1)
19 try:
20     xs = [1, 3, 5, 7, 9, 11]
21     print('trinary_search_go(xs, 7)=', trinary_search_go(xs, 7))
22 except RuntimeError:
23     print('StackOverflow')
24 EOF
25 $ python3 foo.py
```

Fraction of LLMs with correct answer: 16 / 16 = 1.00

Problem 39. Write the output of the final command in the following shell script.

```
1 $ cd; rm -rf quiz; mkdir quiz; cd quiz
2 $ cat > foo.py <<EOF
3 def trinary_search_go(xs, y):
4     def go(lo, hi):
5         if lo > hi:
6             return False
7         mid1 = lo + (hi - lo) // 3
8         mid2 = lo + 2 * (hi - lo) // 3
9         if xs[mid1] == y:
10            return True
11        if xs[mid2] == y:
12            return True
13        if y < xs[mid1]:
14            return go(lo, mid1 - 1)
15        if y < xs[mid2]:
16            return go(mid1 + 2, mid2 - 1)
17        return go(mid2 + 1, hi)
18    return go(0, len(xs) - 1)
19 try:
20     xs = [1, 3, 5, 7, 9, 11]
21     print('trinary_search_go(xs, 5)=',trinary_search_go(xs, 5))
22 except RuntimeError:
23     print('StackOverflow')
24 EOF
25 $ python3 foo.py
```

Fraction of LLMs with correct answer: 11 / 16 = 0.69

Note 40. *Mutual recursion* is when two (or more) functions call each other, forming a cycle of recursive calls. The base case of a mutually recursive pair may live in only one of the functions, so long as every call into the other function is eventually driven back to that base case.

Problem 41. Write the output of the final command in the following shell script.

```
1 $ cd; rm -rf quiz; mkdir quiz; cd quiz
2 $ cat > foo.py <<EOF
3 def sum_even(xs):
4     if len(xs) == 0:
5         return 0
6     return xs[0] + sum_odd(xs[1:])
7
8 def sum_odd(xs):
9     if len(xs) == 0:
10        return 0
11    return xs[0] + sum_even(xs[1:])
12 try:
13     print('sum_even([1, 2, 3, 4, 5])=', sum_even([1, 2, 3, 4, 5]))
14 except RuntimeError:
15     print('StackOverflow')
16 EOF
17 $ python3 foo.py
```

Fraction of LLMs with correct answer: 15 / 16 = 0.94

Problem 42. Write the output of the final command in the following shell script.

```
1 $ cd; rm -rf quiz; mkdir quiz; cd quiz
2 $ cat > foo.py <<EOF
3 def is_even(n):
4     if n == 0:
5         return True
6     return is_odd(n - 1)
7
8 def is_odd(n):
9     return not is_even(n)
10 try:
11     print('is_even(5)=', is_even(5))
12     print('is_odd(5)=', is_odd(5))
13 except RuntimeError:
14     print('StackOverflow')
15 EOF
16 $ python3 foo.py
```

Fraction of LLMs with correct answer: 15 / 16 = 0.94

Problem 43. Write the output of the final command in the following shell script.

```
1 $ cd; rm -rf quiz; mkdir quiz; cd quiz
2 $ cat > foo.py <<EOF
3 def sum_even(xs):
4     if len(xs) == 0:
5         return 0
6     return xs[0] + sum_odd(xs[1:])
7
8 def sum_odd(xs):
9     return xs[0] + sum_even(xs[1:])
10 try:
11     print('sum_even([1, 2, 3, 4, 5])=',sum_even([1, 2, 3, 4, 5]))
12 except RuntimeError:
13     print('StackOverflow')
14 EOF
15 $ python3 foo.py
```

Fraction of LLMs with correct answer: 14 / 16 = 0.88

Problem 44. Write the output of the final command in the following shell script.

```
1 $ cd; rm -rf quiz; mkdir quiz; cd quiz
2 $ cat > foo.py <<EOF
3 def is_even(n):
4     if n == 0:
5         return True
6     return is_odd(n - 1)
7
8 def is_odd(n):
9     if n == 0:
10        return True
11    return is_even(n - 1)
12 try:
13    print('is_even(5)=',is_even(5))
14    print('is_odd(5)=',is_odd(5))
15 except RuntimeError:
16    print('StackOverflow')
17 EOF
18 $ python3 foo.py
```

Fraction of LLMs with correct answer: 11 / 16 = 0.69

LLM Model Performance

